MIL-P-46222E (PA)
10 July 1975
SUPERSEDING
MIL-P-46222D (MU)
12 August 1969

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M80A1
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Electric, M80A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-B-45451 - Benite Strands
MIL-I-46224 - Igniter, Electric, Primer, M63, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes, (ABC-STD-105).

FSC: 1390

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants, Solid: Sampling, Examination and testing
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8837899 - Carton, Packing, Ammunition, for Primer, Electric, M80A1
8837900 - Box, Packing, Ammunition, for Primer, Electric, M80A1
8839499 - Primer, Electric, M80A1, Loading Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primers.-The loaded primers shall comply with all requirements specified on Drawing (dwg.) 8839499 and with all requirements specified in applicable specifications.

3.3 Moisture content of benite.-The moisture content of the benite at the loading station at the time of loading shall not exceed 1.00 percent when tested as specified in 4.3.1.

3.4 Electrical continuity.-The primers shall have an electrical resistance of not less than 0.80 ohm nor greater than 1.80 ohms when tested as specified in 4.3.2.

3.5 Lacquer in flash holes.-The lacquer in primer flash holes shall comply with illustration 1, 2 or 3 of Figure 1.

3.6 Proving ground.-The primer shall comply with the following requirements when tested as specified in 4.3.3.

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.

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c. There shall be no audible hangfires.
d. The primer shall not squib.

3.7 Workmanship.-Loading and assembling shall be accomplished in a thorough, workmanlike manner. Each component and assembly shall be free of cracks, dirt, grease, rust, corrosion products or other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Inspection Provisions

4.2.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection

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procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Benite of one lot interfix number from one manufacturer.

c. Igniter of one lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Primer assembly, prior to loading (see dwg. 8839499).

Categories	Defects	Method of Inspection
Critical:		
	1. Any flash hole missing	Gage
Major:		
	101. Component damaged	Visual
	102. Evidence of poor workmanship (see 3.7)	Visual
Minor: None defined		

4.2.2.2 Charge, prior to loading (see dwg. 8839499)

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:		
	101. Weight of charge	Balance
	102. Length of charge	Gage

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Minor:
201. Foreign matter in charge .....Visual

4.2.2.3 Primer assembly, prior to assembly of plug (see dwg. 8839499)

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Lacquer missing from plug threads	Visual
	102. Charge missing	Visual
	103. Liner improperly positioned	Visual
Minor:	None defined.	

4.2.2.4 Primer assembly (see dwg. 8839499)

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Stake missing	Visual
	102. Flash hole sealing improper (see 3.5)	Visual
	103. Split or cracked head	Visual
	104. Igniter above flush with surface of head	Visual-Manual
Minor:		
	201. Total length, maximum (max.)	Gage CD
	202. Plug above flush with end of body	Visual
	203. Threads damaged	Visual
	204. Marking missing, incorrect or unidentifiable	Visual
	205. Evidence of poor workmanship (see 3.7)	Visual

4.2.2.5 Carton, prior to sealing (see dwg. 8837899)

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
	101. Any packing material missing	Visual

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102. Separator damaged to the extent that primers touch each other ....Visual

Minor: None defined

4.2.2.6 Carton, sealed (see dwg. 8837899)

Categories	Defects	Method of Inspection
Critical: None defined		
Major: None defined.		
Minor:		
201.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton completely	Visual
202.	Carton torn, cut or punctured	Visual
203.	Contents loose to an extent greater than permitted by packing design	Manual
204.	Marking missing, incorrect or unidentifiable	Visual

4.2.2.7 Sealed barrier bag (see dwg. 8837899)

Categories	Defects	Method of Inspection
Major:		
101.	Bag punctured, torn or cut	Visual
102.	Seal mislocated or incomplete	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.7)	Visual

4.2.2.8 Box, packing prior to sealing (see dwg. 8837900)

Categories	Defects	Method of Inspection
Critical: None defined		

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Major:
101. Carton missing .....Visual

Minor: None defined

4.2.2.9 Box packing, sealed (see dwg. 8837900)

Categories	Defects	Method of Inspection
Critical: None defined		
Major:		
101.	Protruding nail	Visual
102.	Board broken or loose	Visual
103.	Strapping missing, broken or loose	Visual
Minor:		
201.	Contents loose	Manual
202.	Strapping improperly engaged or mislocated	Visual
203.	Marking missing, incorrect or unidentifiable	Visual
204.	Evidence of poor workmanship (see 3.7)	Visual

4.2.3 Testing

4.2.3.1 Moisture content of benite (see 3.3) Major defect.- The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. For verification, one 100 gram sample of benite shall be selected at the loading station, during each eight hours production. If the moisture content of the sample exceeds the applicable requirement the primers loaded since the last acceptable sample shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 Electrical continuity (see 3.4) Major defect - This test shall be performed 100 percent as specified in 4.3.2 using the applicable equipment in accordance with 4.2.4. Any unit exhibiting a resistance outside the permitted range shall be classed defective and removed from the lot.

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4.2.3.3 Functioning (see 3.6).-The primer assemblies shall be observed for evidence of the following defects:

Table I

Defects	Classification
Body ruptures or becomes detached	Critical
Failure to fire	Major
Assembly hangfires	Major
Assembly squibs	Major

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 35 primers shall be randomly selected from each lot for this test. If a critical defect occurs or if two or more major defects occur, the lot shall be rejected.

4.2.3.3.2 After three consecutive lots have met the criteria specified in 4.2.3.3.1, 10 primers shall be randomly selected from each lot. If a critical defect occurs or if one or more major defects occur, the lot shall be rejected.

4.2.3.4 Check test for deterioration of Igniters, Electric M63, Major defect, (see 6.3).-If the igniters have been stored under normal conditions more than 2 years or for any period of time under adverse conditions, however brief, since previous tests, the igniter lot shall be subject to, and must comply with, the check test specified in Specification MIL-I-46224 immediately before assembly into the primers of this specification.

4.2.3.5 Heat seal test of barrier bag (see dwg. 8837899) (Major defect).-Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in the section is identified under the "Method of Inspection" headlines in 4.2.2 and 4.2.3 (see 6.2).

4.3 Test methods and procedures

4.3.1 Moisture content of benite.-The moisture content of the benite shall be tested in accordance with Method 102.1 of Standard MIL-STD-286.

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4.3.2 Electrical continuity.-The resistance of the primer shall be taken between the electrode as one contact and the primer head as the other. The applied current shall be limited to 20 milliamperes, maximum. Any unit exhibiting a resistance outside the permitted range shall be classed defective.

4.3.3 Proving ground.-Testing shall be performed at a Government proving ground. The sample primers shall be assembled into complete rounds for which the primer was designed and fired from a weapon for which the round is standard. Observation shall be made for correct velocity level and for compliance with the requirements of 3.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-Packaging shall be in accordance with dwg. 8837899.

5.2 Packing

5.2.1 Level C.-The packaged primers shall be packed in accordance with dwg. 8837900.

5.3 Marking.-Exterior containers shall be marked in accordance with dwg. 8837900.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4. Design responsibility for

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all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-24 "Inspection equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor designs to gether with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T Dover, New Jersey 07801.

6.3 Check test for possible deterioration of primers.-The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of M80A1 Primers which have been stored for more than two years or which have been sujected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the M80A1 Primer is assembled, does not require the performance of the following check test, the contracting officer shall insert the provisions for the check test in the contract.

6.3.1 Moisture content of benite.-A sample of 25 primers shall be selected for this test. The benite shall be removed from the primer by any suitable means. The mositure content of the benite shall be determined in accordance with Standard MIL-STD-286, Method 102.1. If one or more primers fail to comply with the requirement specified in 3.3, the lot shall be rejected.

6.3.2 Electrical resistance.-This test shall be performed 100 percent. Any primer assembly which fails to comply with the requirement specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.2.

6.3.3 Functioning (Igniter, Electric, M63).-300 primers shall be selcted from each lot for this test. The igniters shall be removed from the primers and tested as specified in Specification MIL-I-46224. If three or more igniters fail to comply with the applicable requirement the lot shall be rejected. At the option of the contractor, complete M80A1 primer assemblies can be tested as specified in Specification MIL-I-46224. Observation

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shall be made on functioning of the complete primer assembly. If three or more primers fail to comply with the applicable requirement the lot shall be rejected.

6.3.4 Non-Functioning (Igniter, Electric, M63).-50 primers shall be selected from each lot for this test. The igniters shall be removed from the primers and tested as specified in Specification MIL-I-46224. If one or more igniters fail to comply with the applicable requirement the lot shall be rejected. At the option of the contractor, complete M80A1 primer assemblies can be tested as specified in Specification MIL-I-46224. Observation shall be made on non-functioning of the complete primer assembly. If one or more primers fail to comply with the applicable requirement the lot shall be rejected.

6.4 Cost of check tests.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The tests shall be conducted at Government expense without cost to the contractor who loaded the igniters or to the contractor assembling the igniters into the artillery primers and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the igniter or artillery primer lot was under jurisdiction of either contractor.

6.5 Submission of firing and test data.-In addition to the normal distribution of records for each lot of primers, when the item is procured by the Department of the Army, one copy should be sent to:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: SARPA-QA-A-A (1 copy) and SARPA-AD-E-P (1 copy)

6.6 Definitions:

a. Hangfires.- A perceptible delay occurring between the instant of application of energy and detonation.

b. Squib.-A burning of the primer mixture without detonation.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A095

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

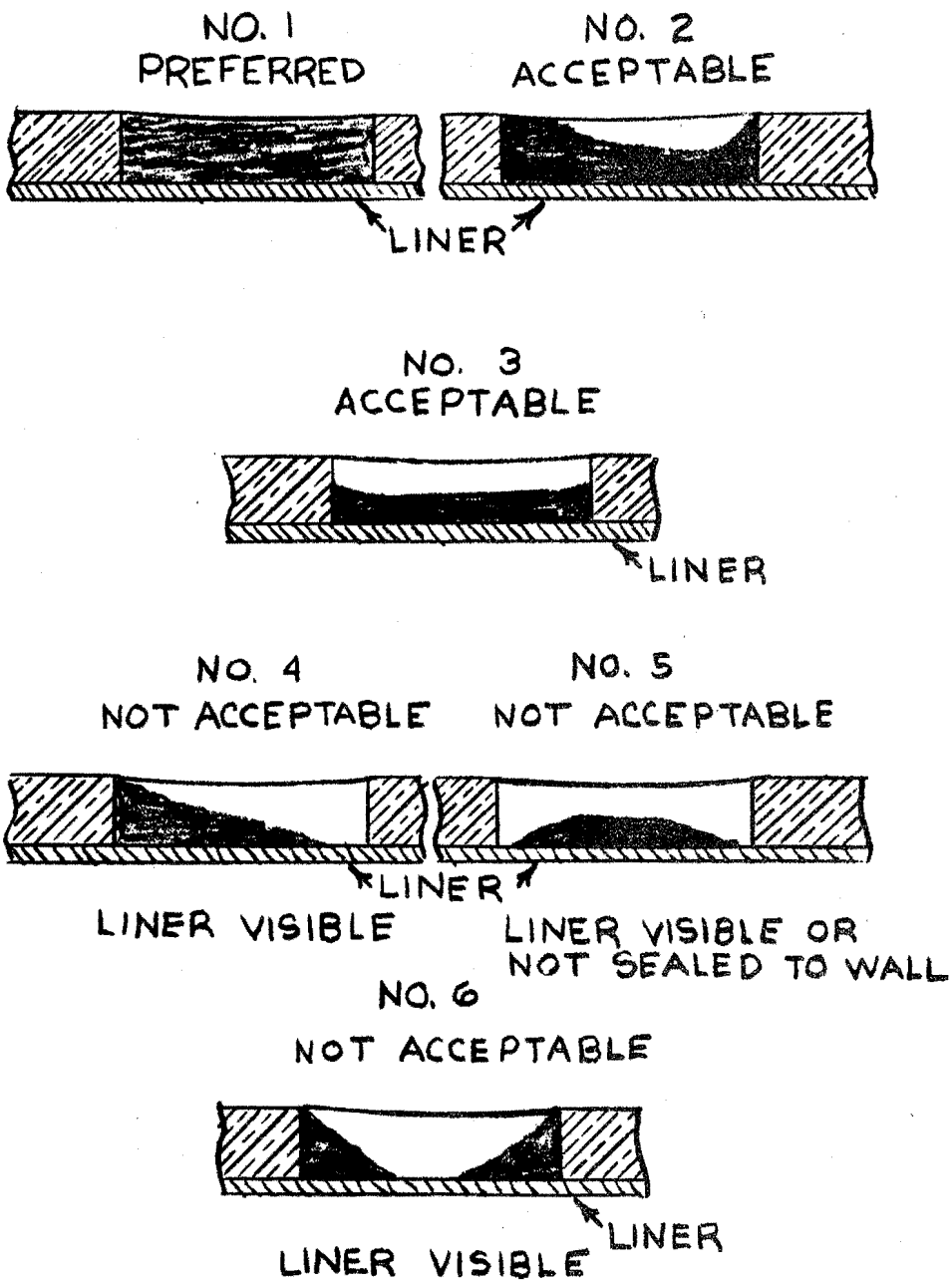


FIGURE 1

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